

Garbage Recognition

Based on the trained model for testing, it can recognize the trained object names. Model misjudgments, non-recognition, and other behaviors are affected by lighting conditions and model parameters. If you want to improve recognition rate, you can collect images for labeling and training yourself.

Main Code

Code path: /home/yahboom/Dofbot/6.AI_Visual/7.Garbage identification.ipynb

The following code content needs to be executed step by step according to actual needs. Do not run all at once. Running the last cell will directly exit the thread. For detailed code, please check the code path.

Import various libraries and model files

```
#!/usr/bin/env python
import Arm_Lib
import os
import cv2 as cv
import threading
from time import sleep
import ipywidgets as widgets
from IPython.display import display
from dofbot_utils.fps import FPS
from ultralytics import YOLO
from dofbot_utils.robot_controller import Robot_Controller

yolo_model = YOLO("./best.pt", task='detect')
fps = FPS()
```

Garbage name list:

```
def garbage_getName(self):
    name = "None"
    if self.status == 'waiting':
        self.frame, msg = self.garbage_identify.garbage_run(self.frame)
        for key, pos in msg.items(): name = key
        if name == "Zip_top_can":
            (self.garbage_num,
self.garbage_class) = ('00', '01')
        if name == "Old_school_bag":
            (self.garbage_num,
self.garbage_class) = ('01', '01')
        if name == "Newspaper":
            (self.garbage_num,
self.garbage_class) = ('02', '01')
        if name == "Book":
            (self.garbage_num,
self.garbage_class) = ('03', '01')
        if name == "Toilet_paper":
            (self.garbage_num,
self.garbage_class) = ('04', '02')
        if name == "Peach_pit":
            (self.garbage_num,
self.garbage_class) = ('05', '02')
```

```

        if name == "Cigarette_butts":                (self.garbage_num,
self.garbage_class) = ('06', '02')
        if name == "Disposable_chopsticks":          (self.garbage_num,
self.garbage_class) = ('07', '02')
        if name == "Egg_shell":                      (self.garbage_num,
self.garbage_class) = ('08', '03')
        if name == "Apple_core":                    (self.garbage_num,
self.garbage_class) = ('09', '03')
        if name == "Watermelon_rind":                (self.garbage_num,
self.garbage_class) = ('10', '03')
        if name == "Fish_bone":                      (self.garbage_num,
self.garbage_class) = ('11', '03')
        if name == "Expired_tablets":                (self.garbage_num,
self.garbage_class) = ('12', '04')
        if name == "Expired_cosmetics":              (self.garbage_num,
self.garbage_class) = ('13', '04')
        if name == "Used_batteries":                 (self.garbage_num,
self.garbage_class) = ('14', '04')
        if name == "Syringe":                        (self.garbage_num,
self.garbage_class) = ('15', '04')
        if name == "None":                          (self.garbage_num,
self.garbage_class) = ('None', 'None')

```

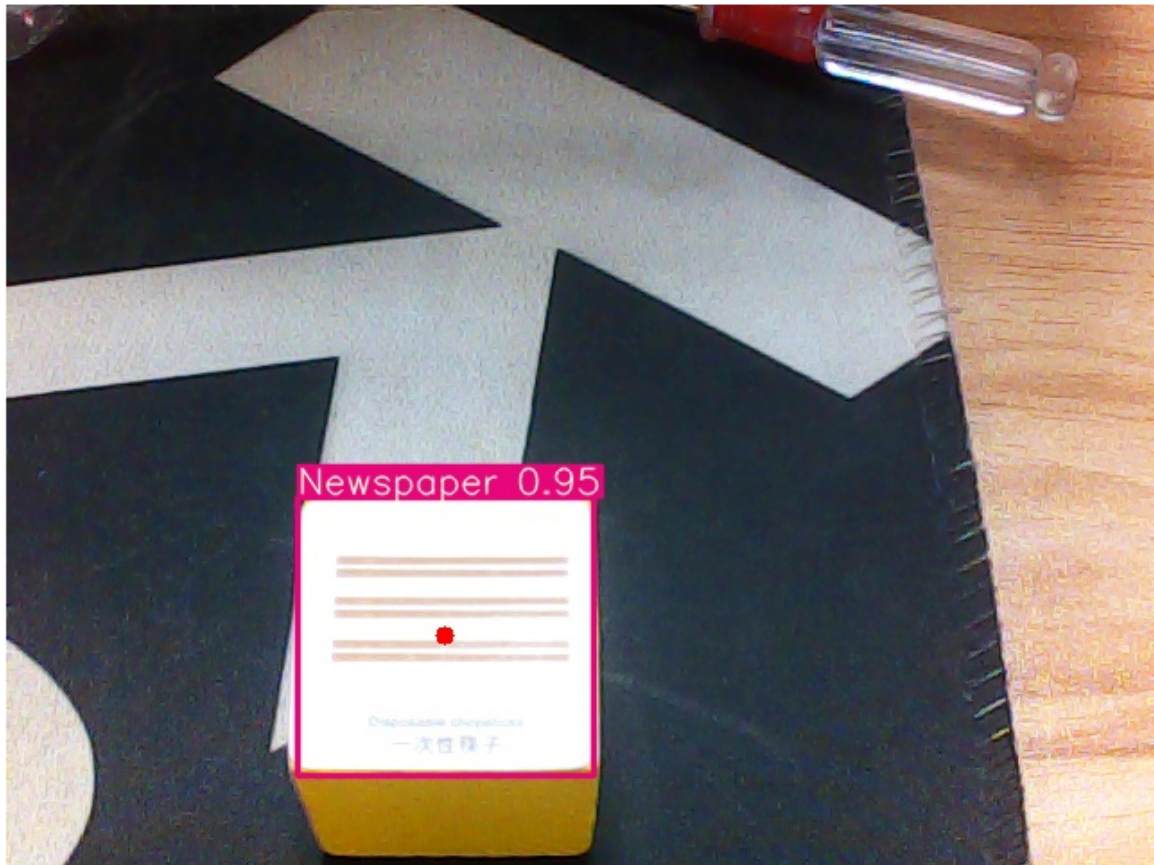
Main thread:

```

def camera():
    # Open camera
    capture = cv.VideoCapture(0)
    capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
    capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)
    # Loop while camera is open
    while capture.isOpened():
        try:
            _, img = capture.read()
            fps.update_fps()
            results = yolo_model(img, save=False, verbose=False) # Use YOLO11
            for object detection
                annotated_frame = results[0].plot(
                    labels = True, # Show labels
                    conf = False, # Show confidence
                    boxes = True, # Draw bounding boxes
                )
            if model == 'Exit':
                cv.destroyAllWindows()
                capture.release()
                break
            fps.show_fps(annotated_frame)
            imgbox.value = cv.imencode('.jpg', annotated_frame)[1].tobytes()
        except Exception as e:
            capture.release()
            print(e)
            break

```

After running the code blocks, you can see the interface below with the corresponding garbage recognized.



Exit